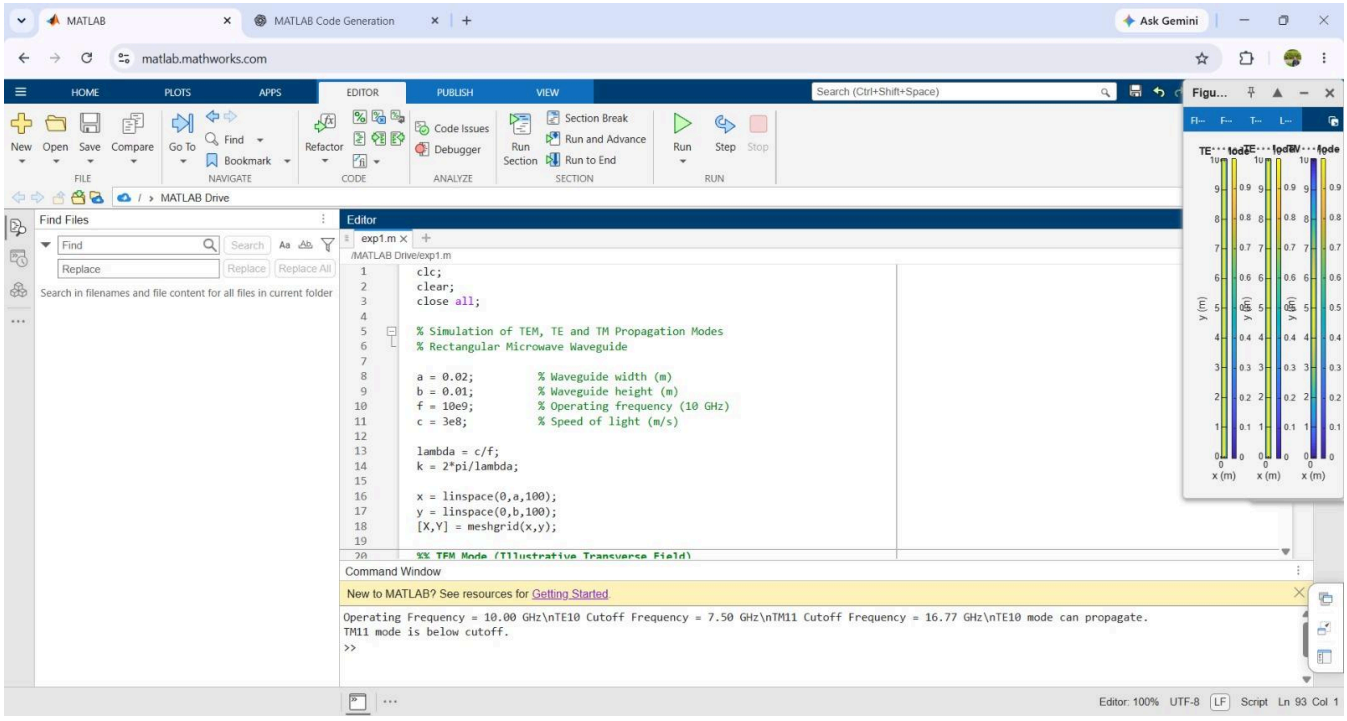
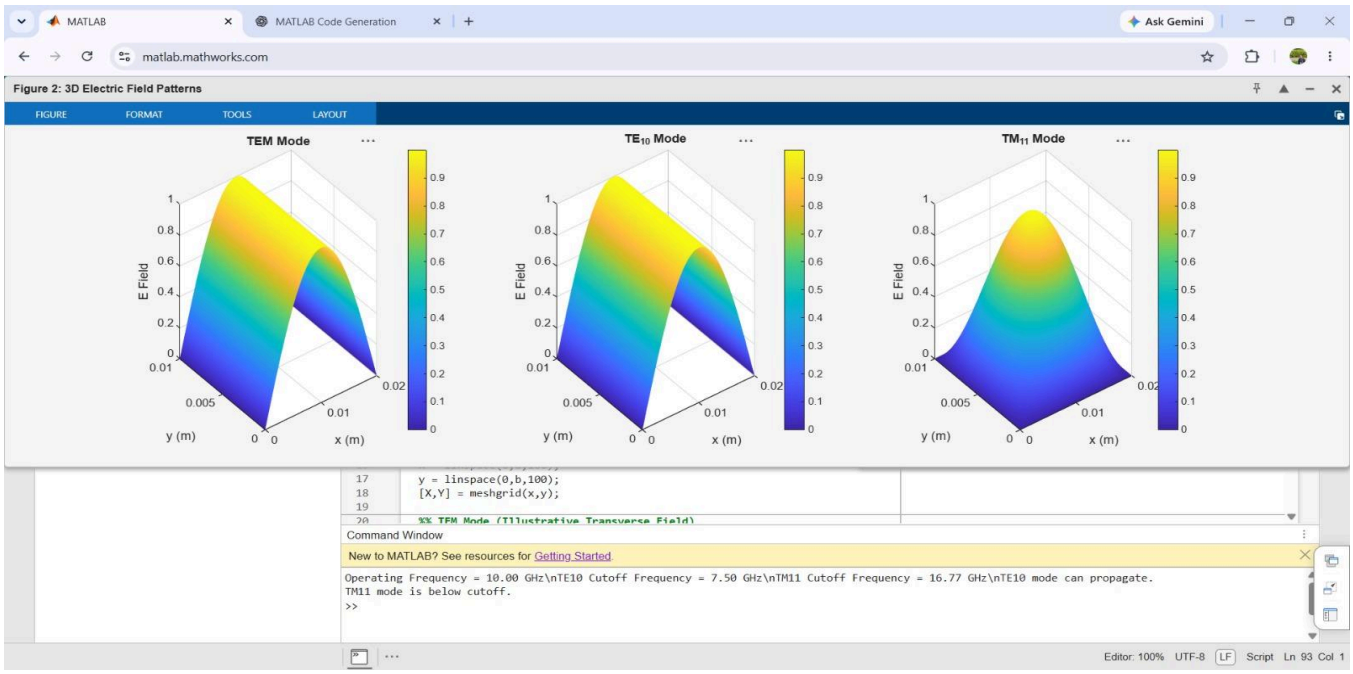
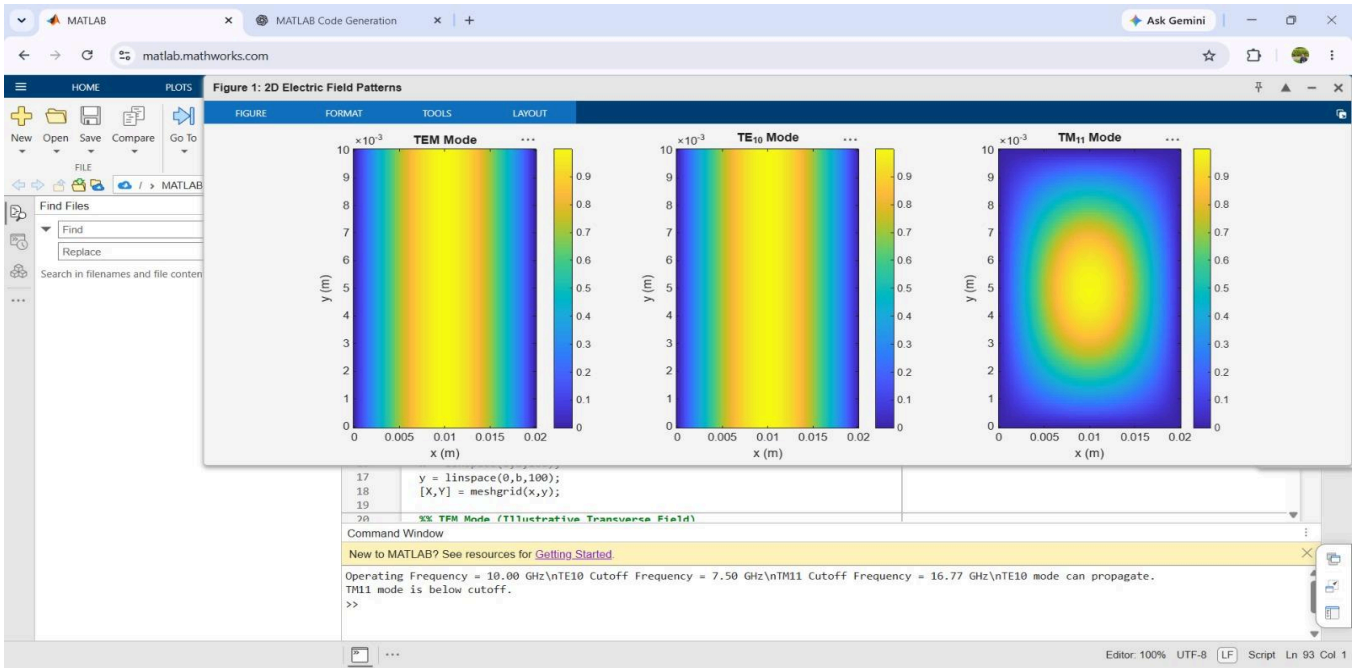
Exp 1: Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

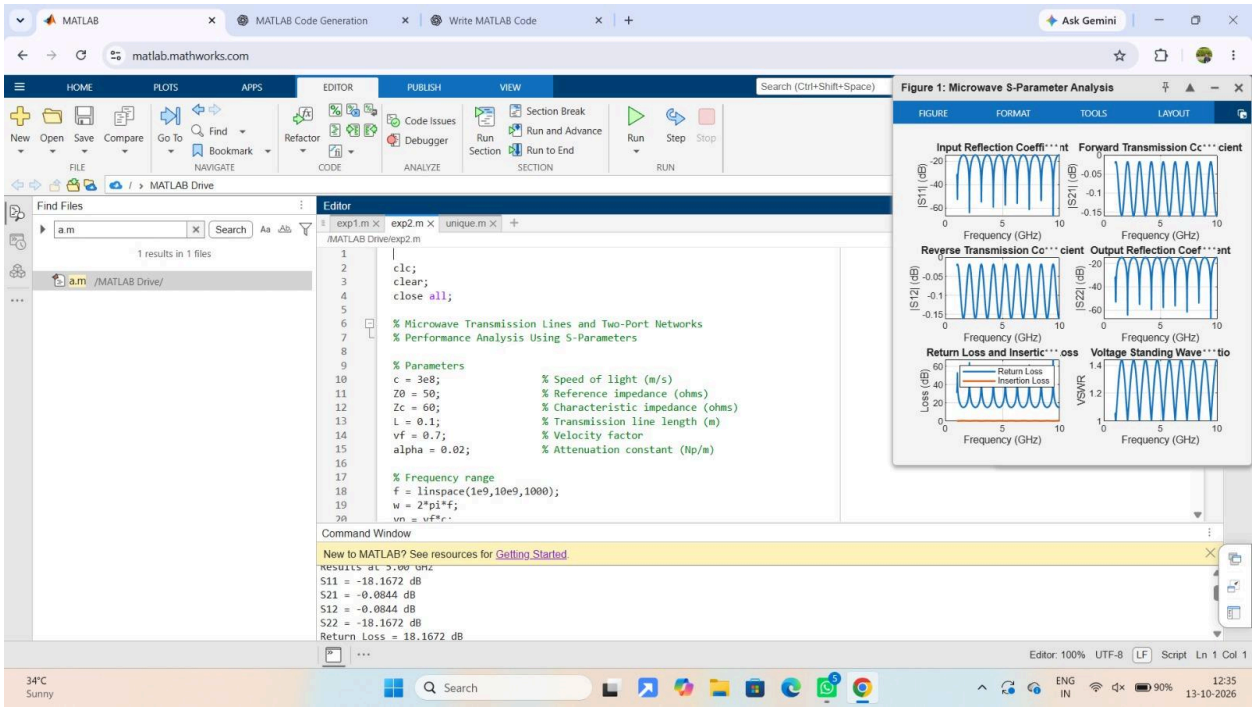


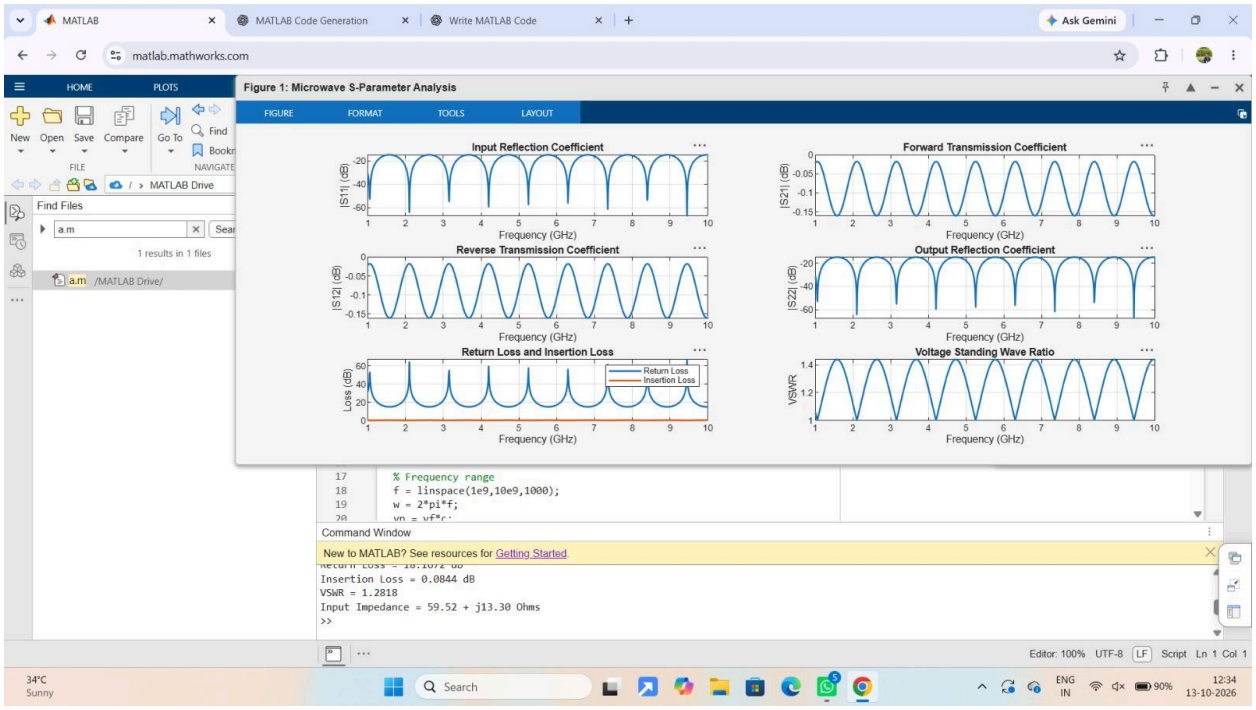
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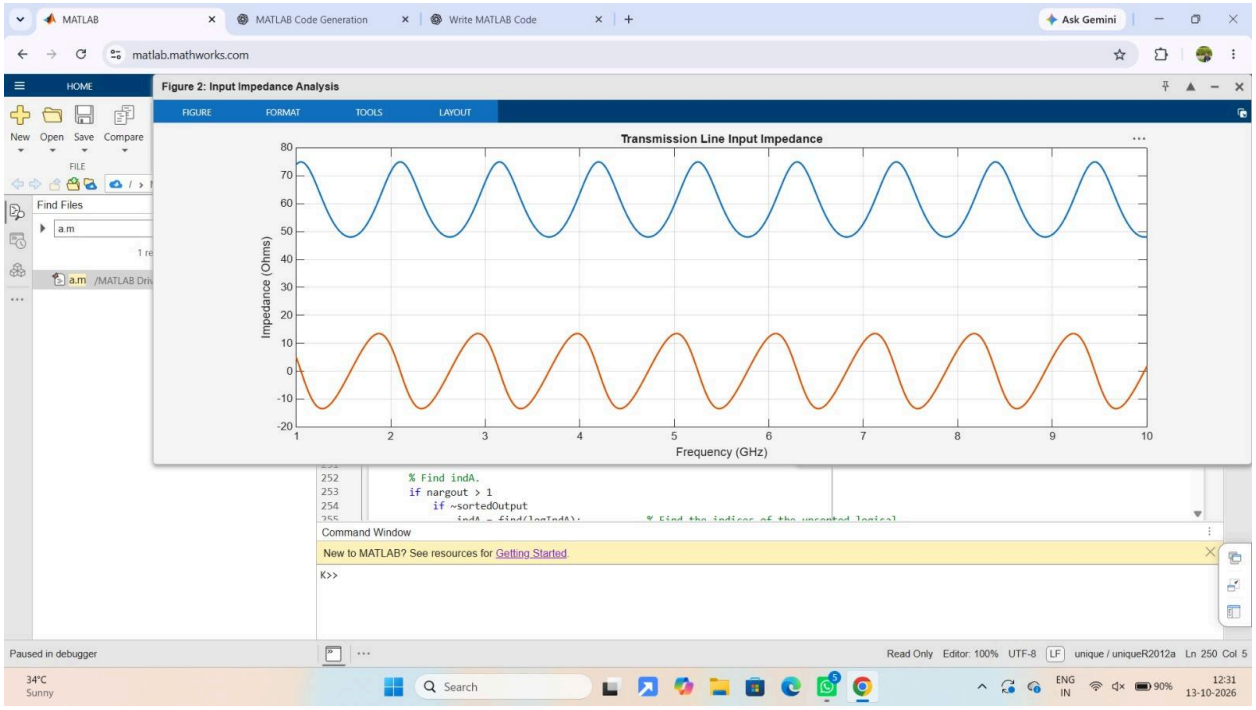




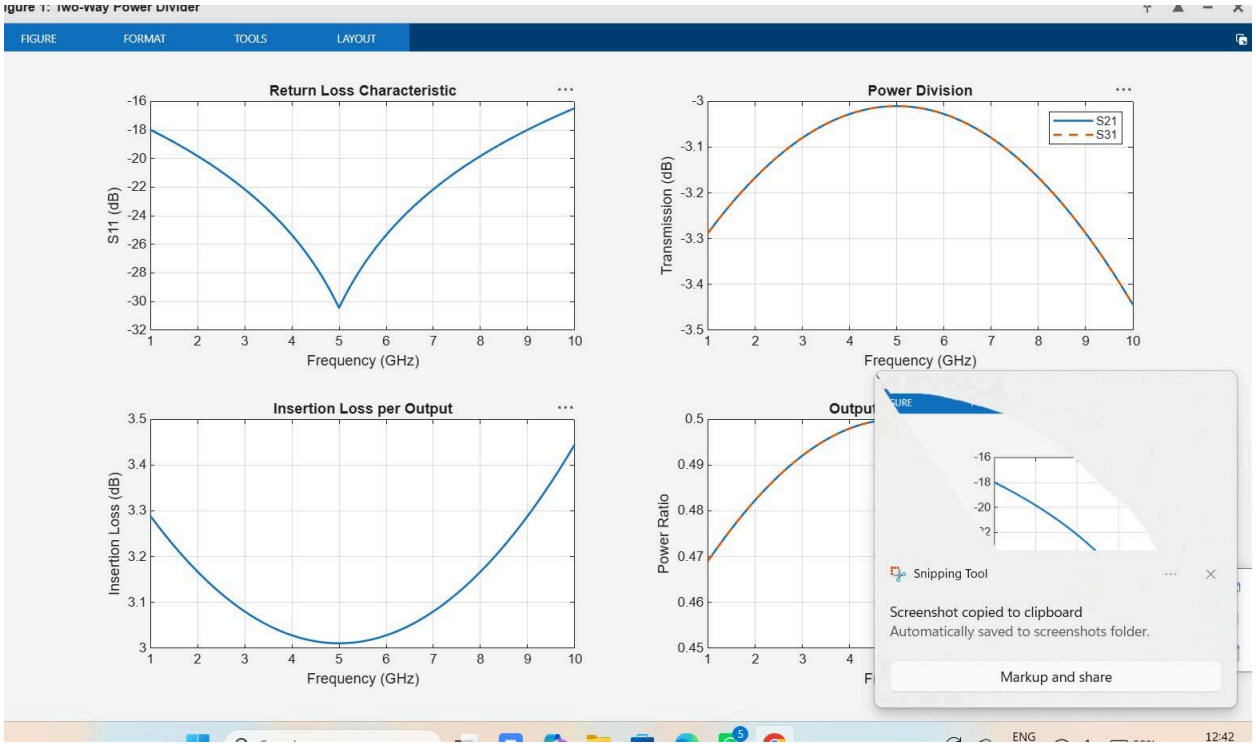
Exp 2:Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

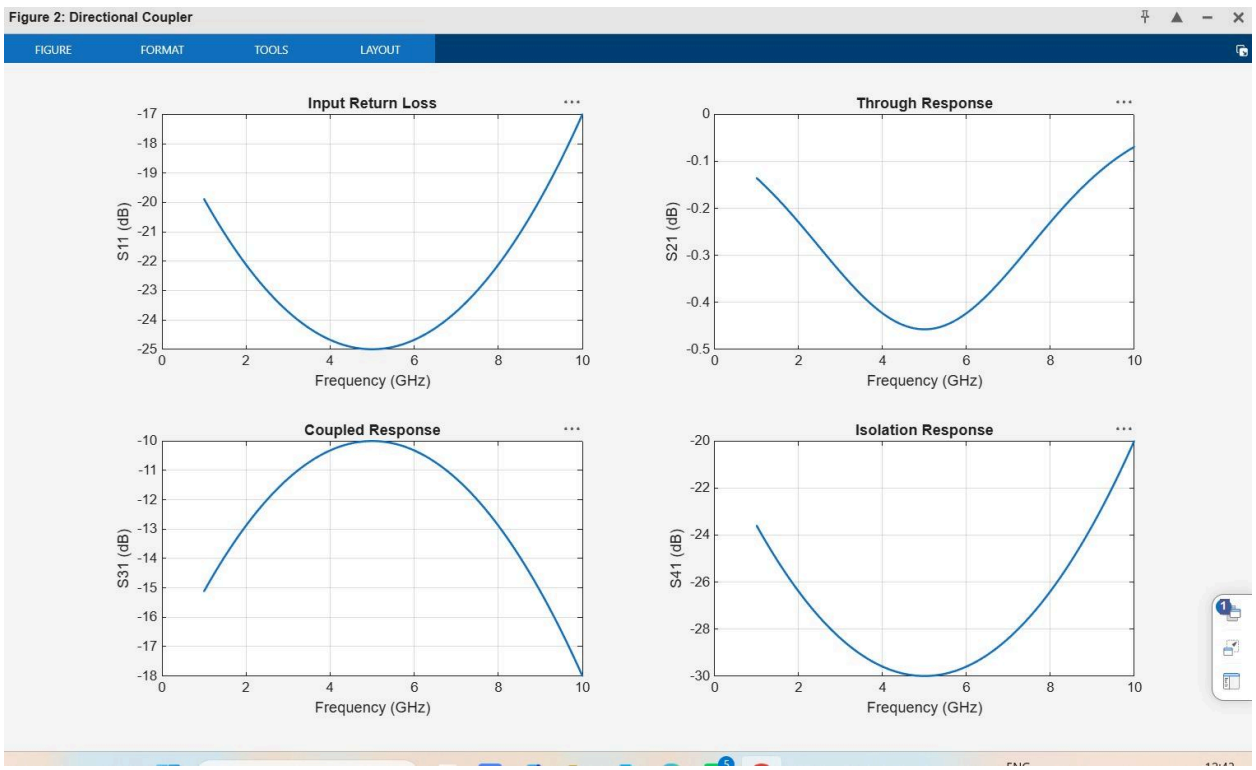



Figure 1: Microwave S-Parameter Analysis
Input Reflection Coefficient
Forward Transmission Coefficient
Reverse Transmission Coefficient
Output Reflection Coefficient
Return Loss and Insertion Loss
Voltage Standing Wave Ratio
Frequency (GHz)
Return Loss
Insertion Loss
% Frequency range
f = linspace(1e9,10e9,1000);
w = 2*pi*f;
Command Window
New to MATLAB? See resources for Getting Started.
Insertion Loss = 0.0844 dB
VSWR = 1.2818
Input Impedance = 59.52 + j13.30 Ohms
>>

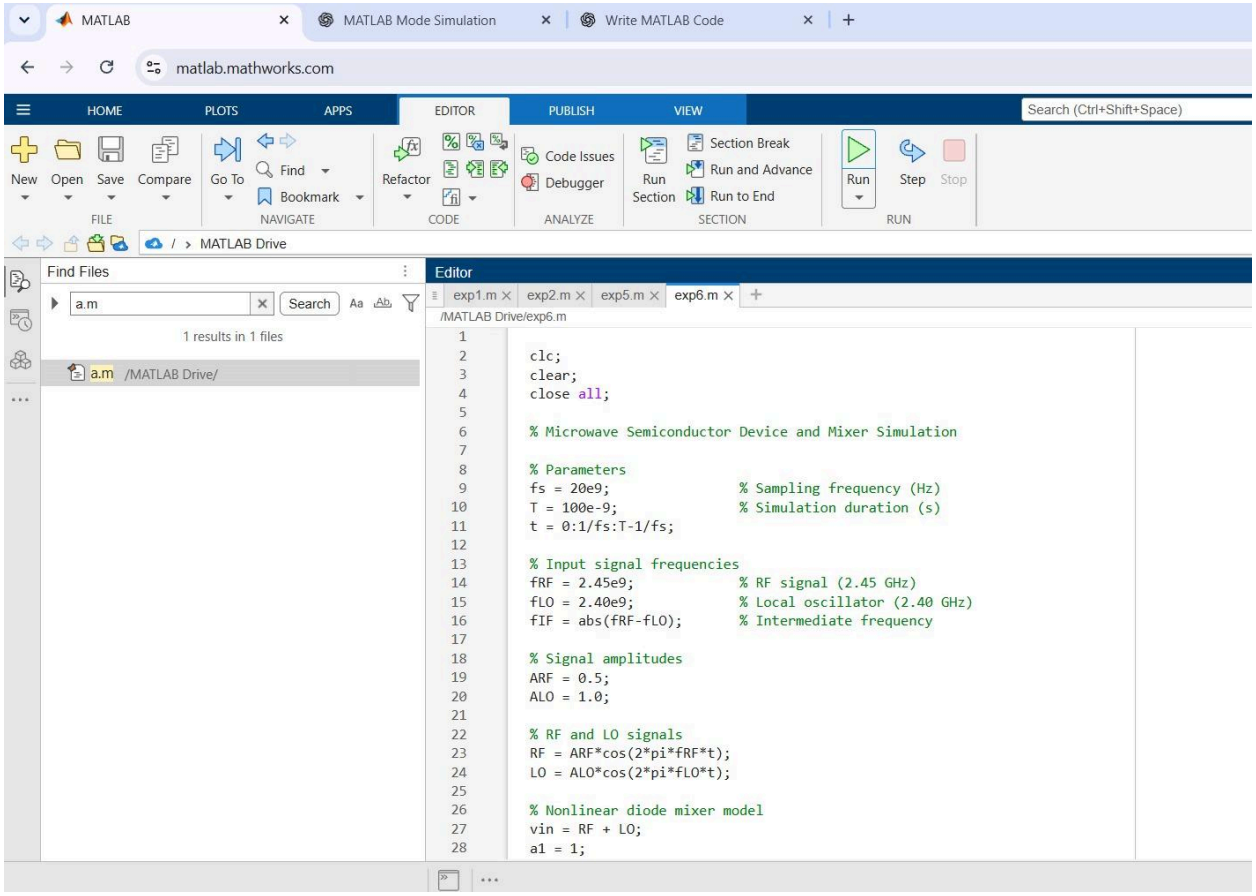

Figure 2: Input Impedance Analysis
Transmission Line Input Impedance
Impedance (Ohms)
Frequency (GHz)
% Find indA.
if nargout > 1
if ~sortedOutput
Command Window
New to MATLAB? See resources for Getting Started.
K>>

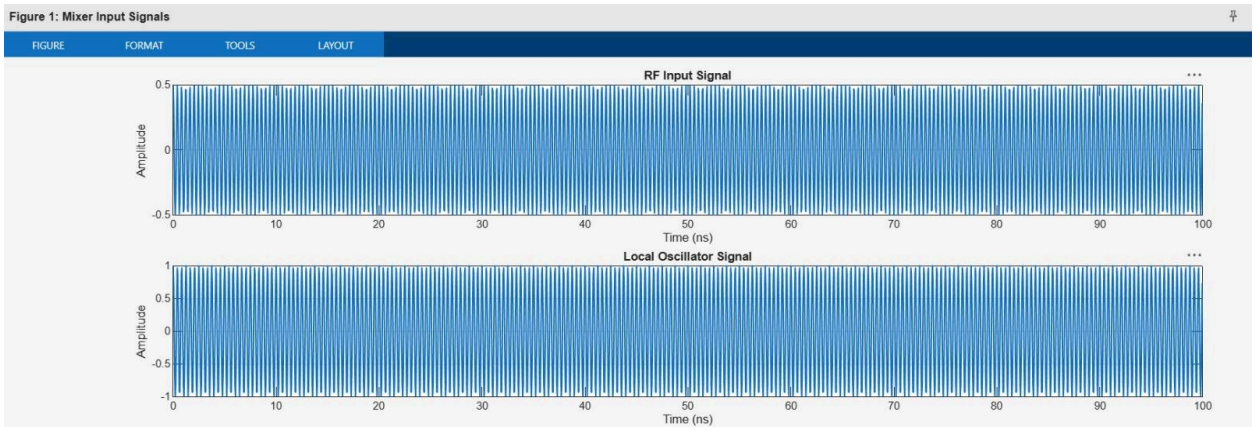
Exp5:Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider

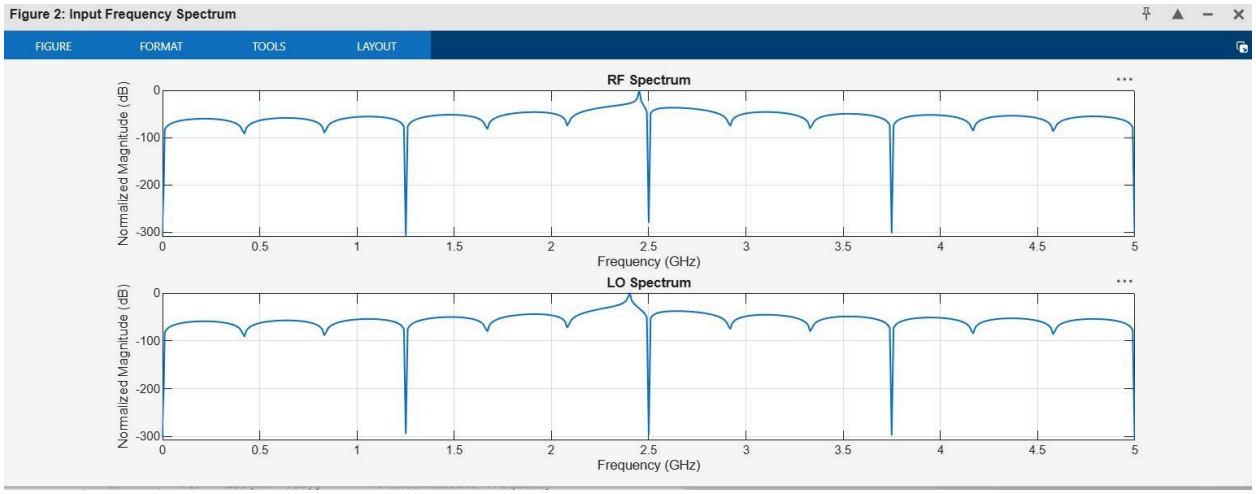


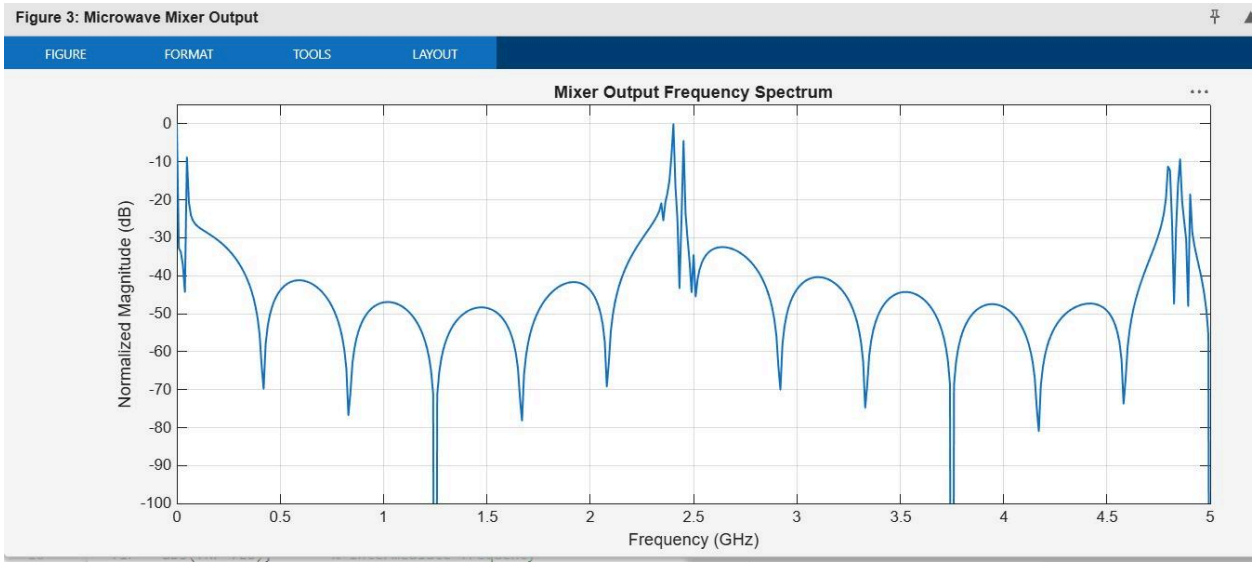


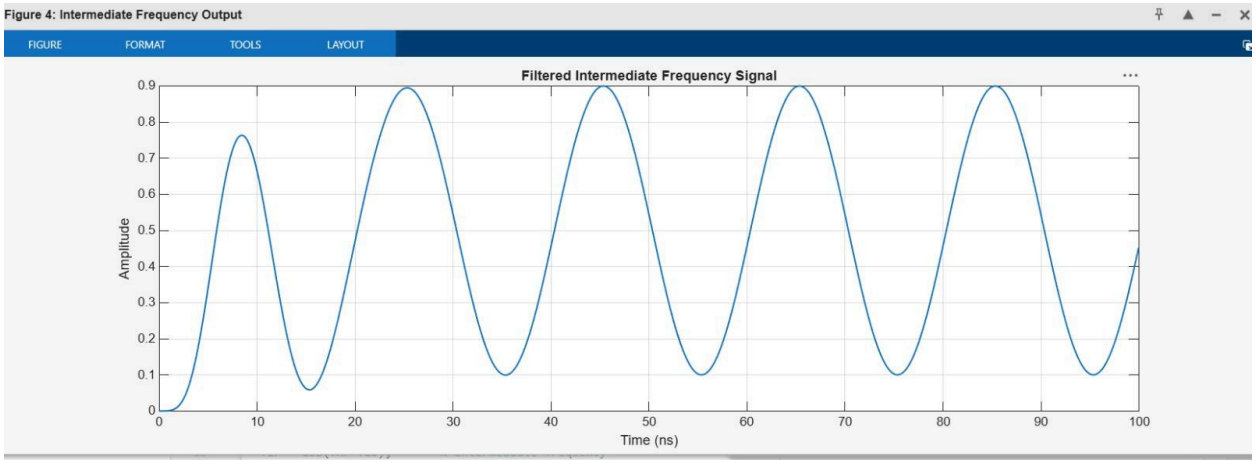
Exp 6:Simulation-Based Investigation of Microwave Semiconductor Devices and Frequency-Conversion Characteristics of a Microwave Mixer



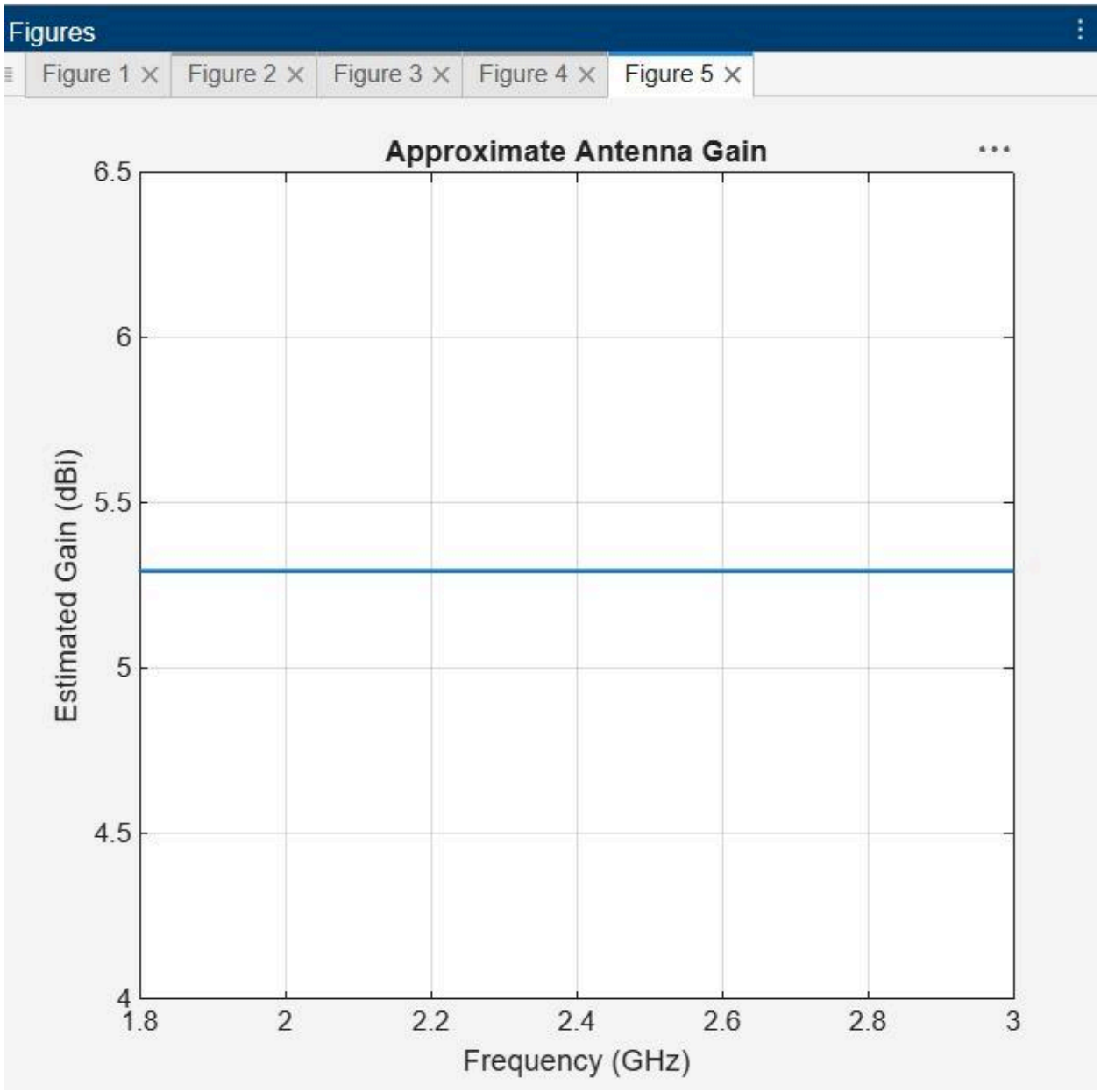


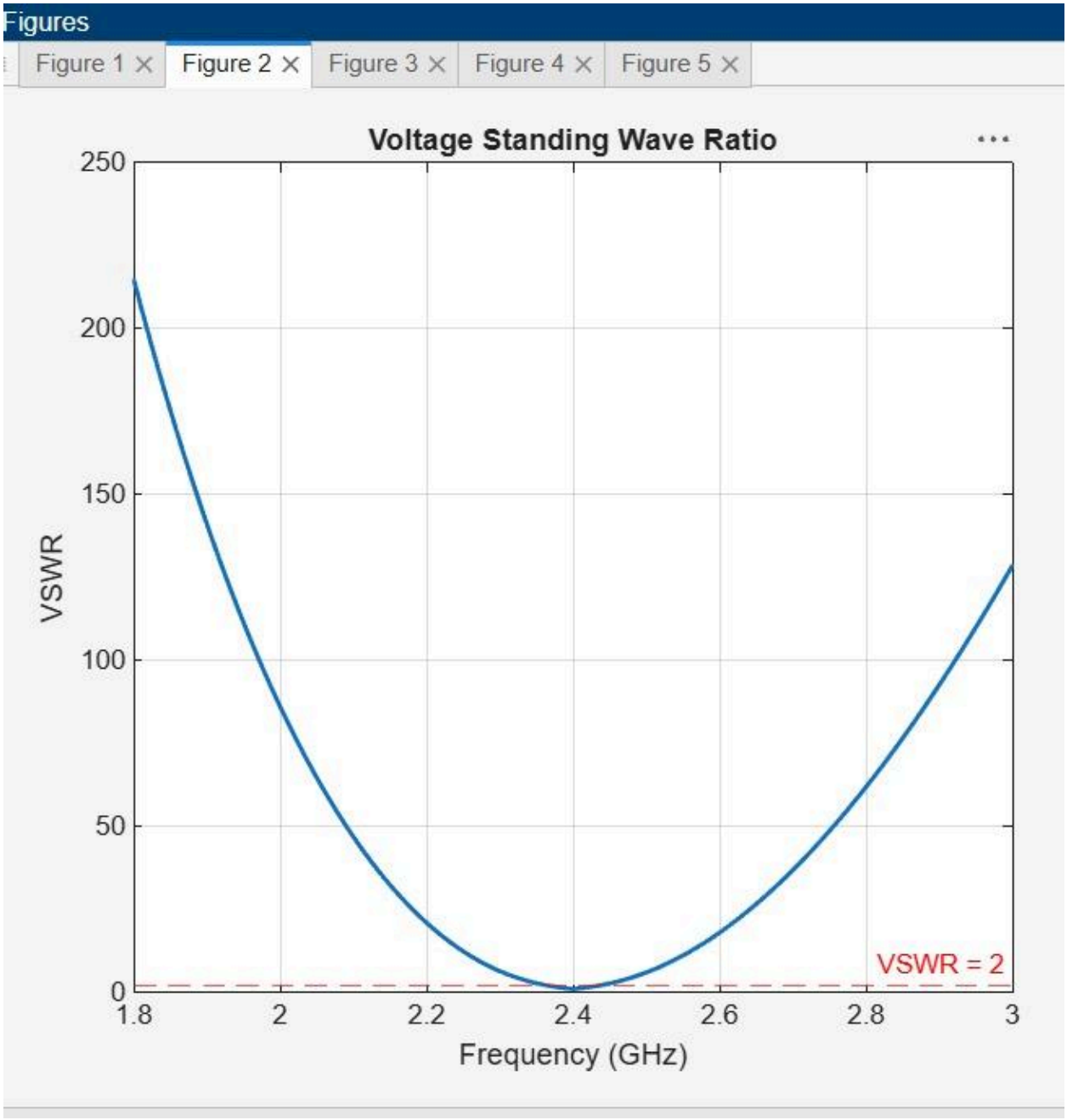

Figure 2: Input Frequency Spectrum
FIGURE
FORMAT
TOOLS
LAYOUT
RF Spectrum
Normalized Magnitude (dB)
0
-100
-200
-300
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Frequency (GHz)
LO Spectrum
Normalized Magnitude (dB)
0
-100
-200
-300
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Frequency (GHz)

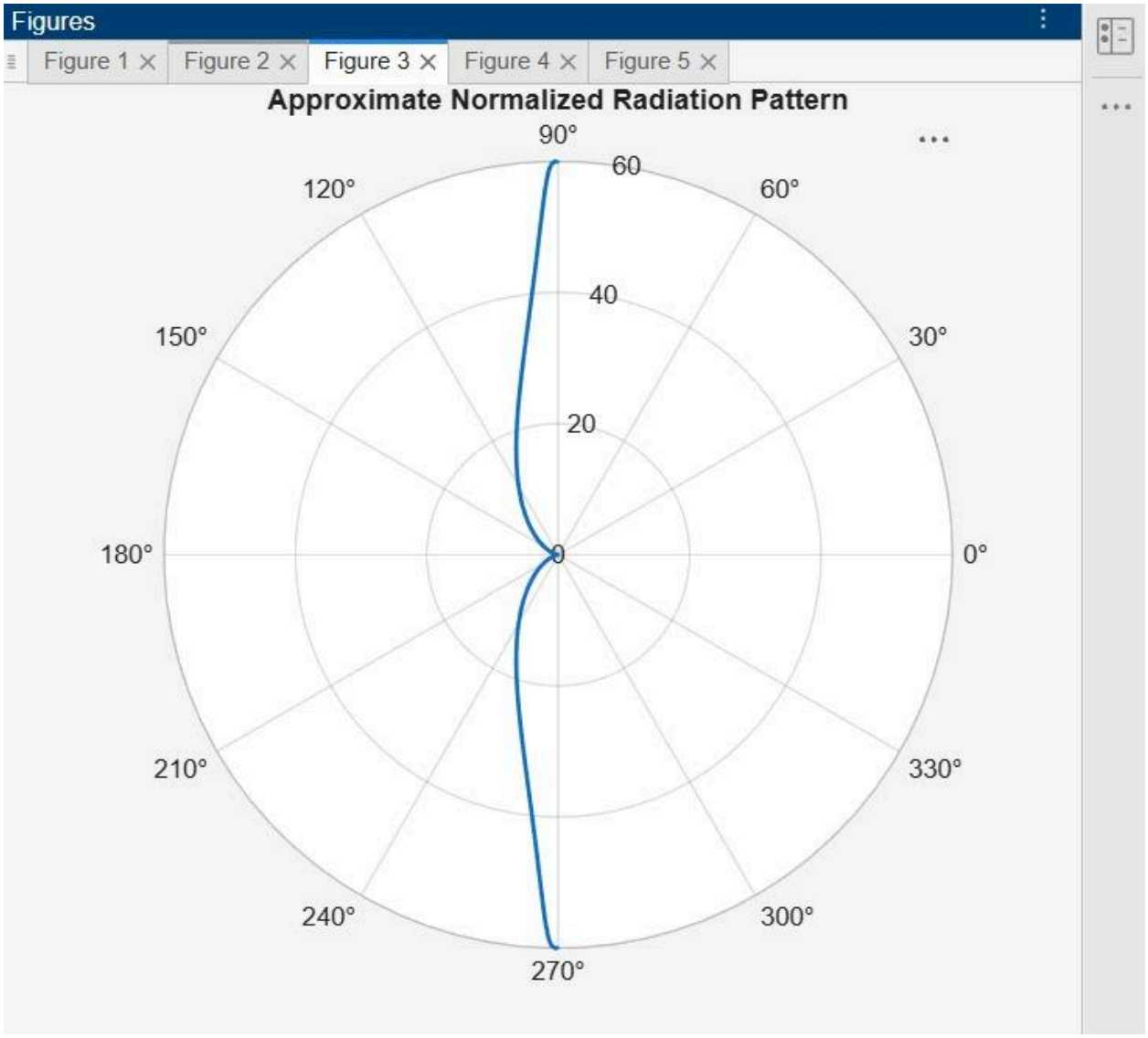

Figure 3: Microwave Mixer Output
FIGURE
FORMAT
TOOLS
LAYOUT
Mixer Output Frequency Spectrum
Normalized Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Frequency (GHz)

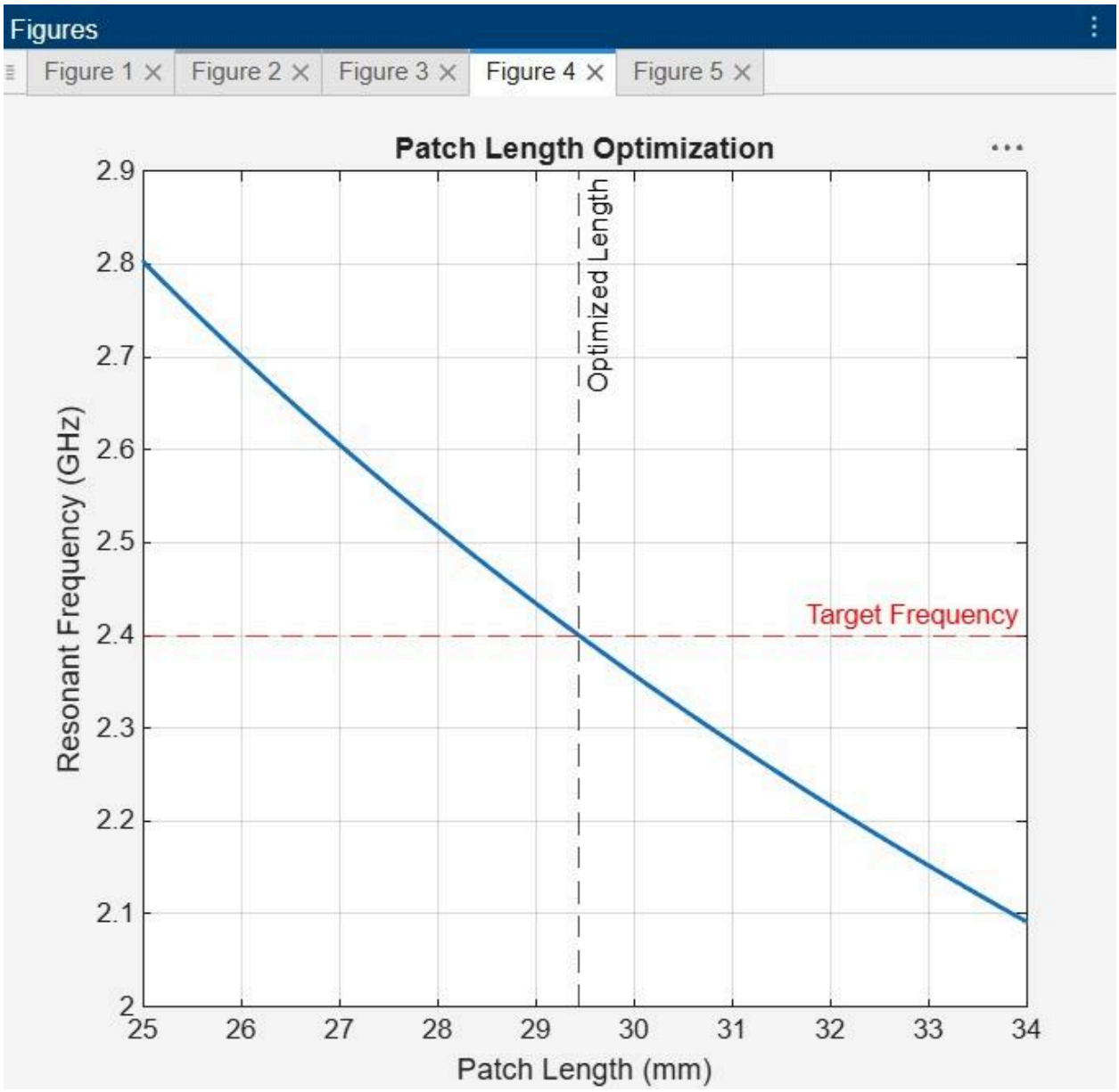

Figure 4: Intermediate Frequency Output
FIGURE
FORMAT
TOOLS
LAYOUT
Filtered Intermediate Frequency Signal
Amplitude
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
10
20
30
40
50
60
70
80
90
100
Time (ns)

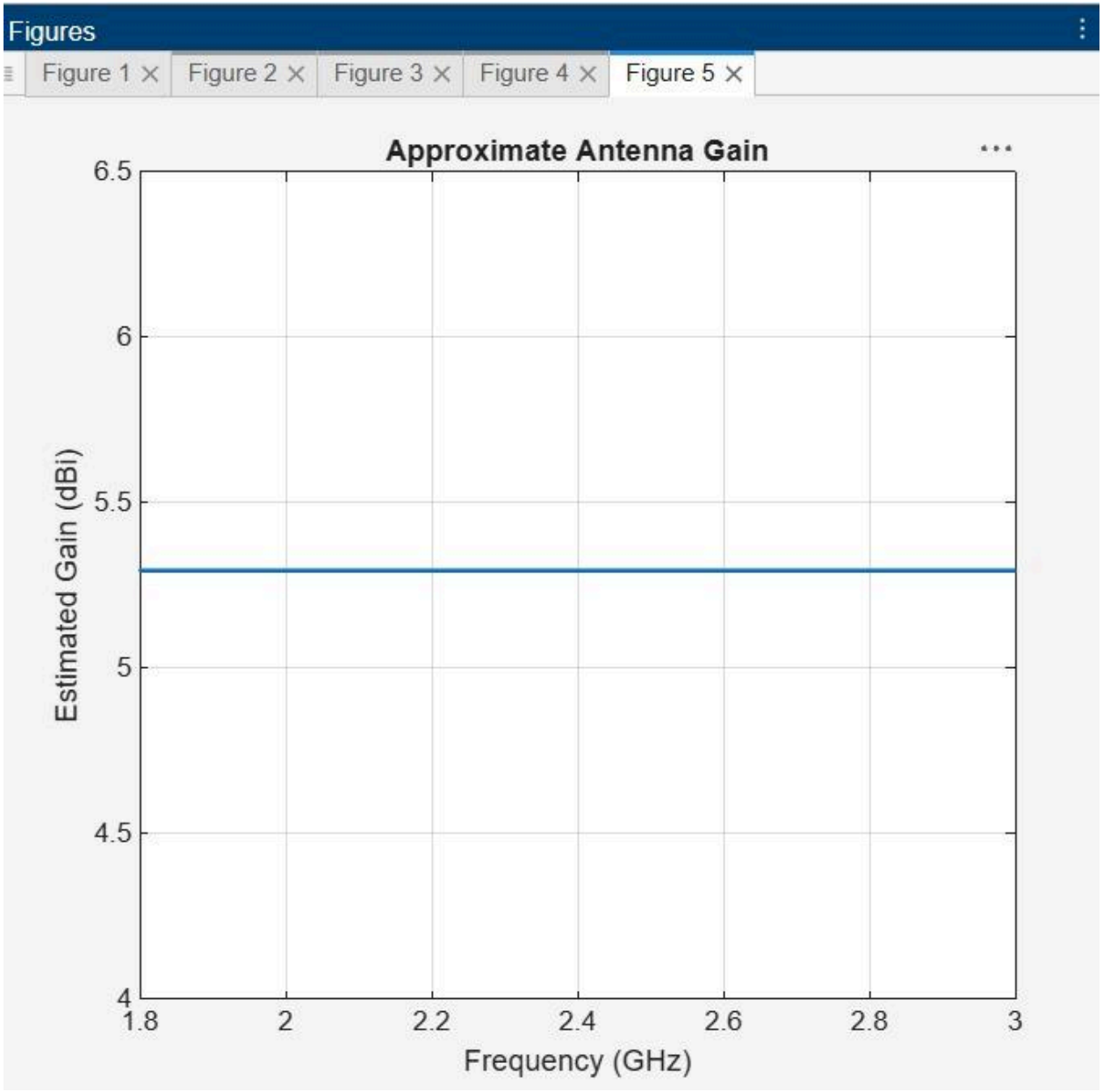
Exp 10:Full-Wave Simulation and Optimization of a Microstrip Patch Antenna for Resonant Frequency, Bandwidth, Gain and Radiation-Pattern Enhancemen



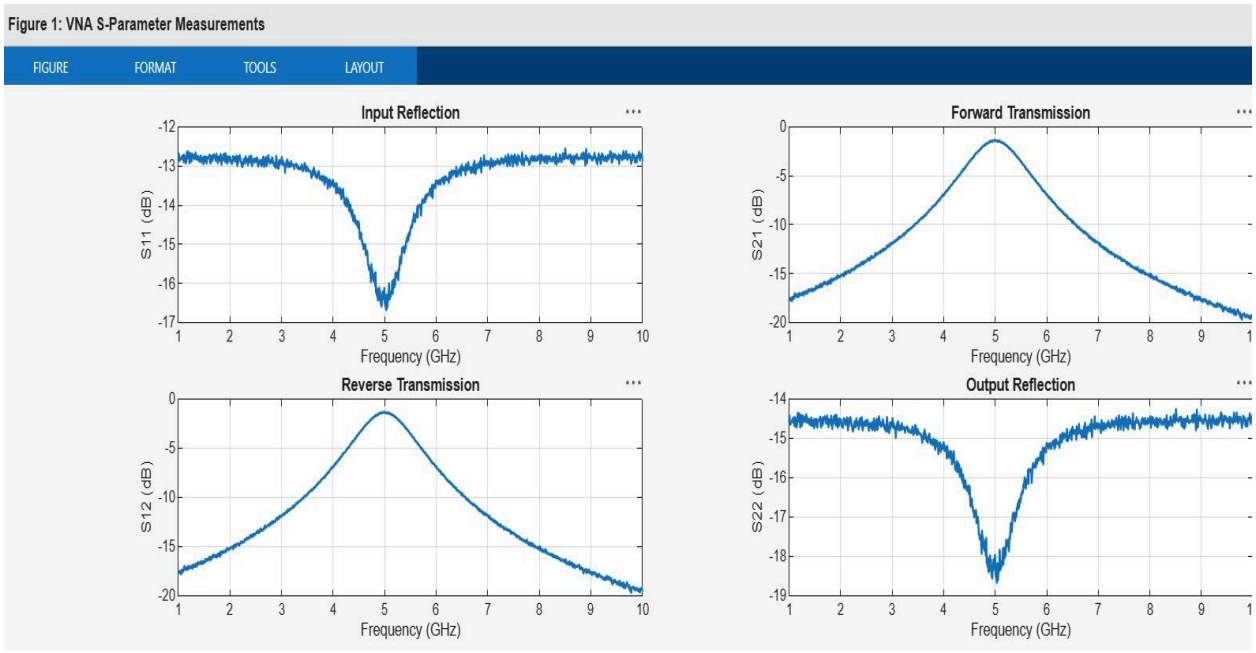

Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Voltage Standing Wave Ratio
VSWR
Frequency (GHz)
VSWR = 2
250
200
150
100
50
0
1.8
2
2.2
2.4
2.6
2.8
3

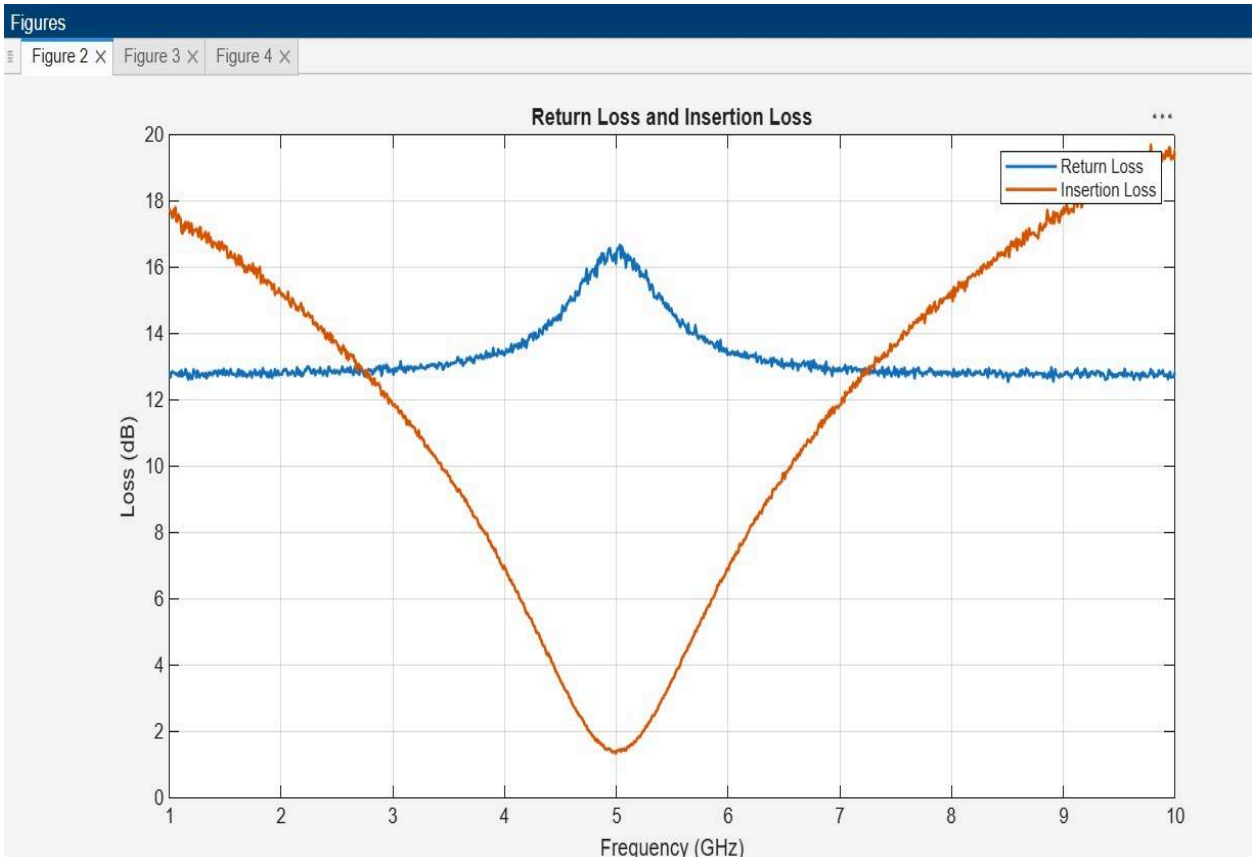

Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Approximate Normalized Radiation Pattern
90°
60
120°
60°
40
150°
30°
20
180°
0°
0
210°
330°
240°
300°
270°

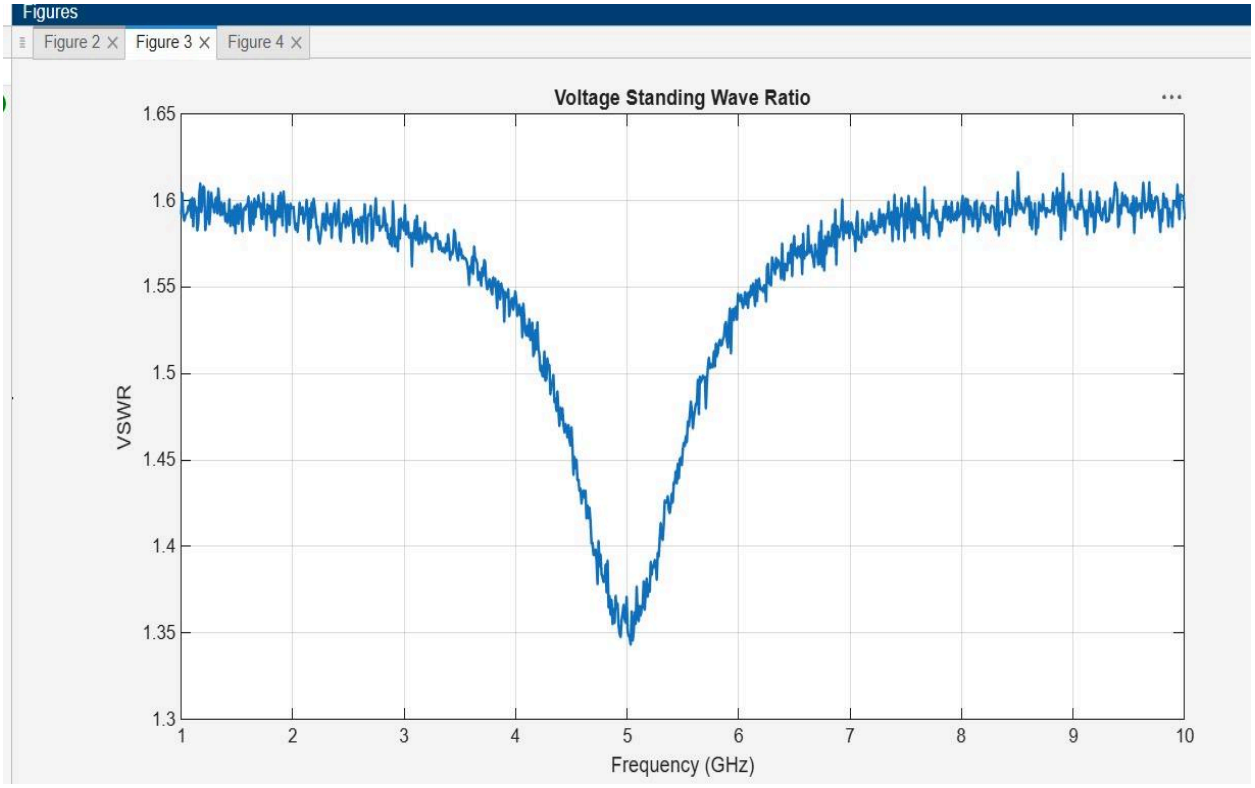

Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Patch Length Optimization
Optimized Length
Target Frequency
Resonant Frequency (GHz)
Patch Length (mm)
2.9
2.8
2.7
2.6
2.5
2.4
2.3
2.2
2.1
2
25
26
27
28
29
30
31
32
33
34

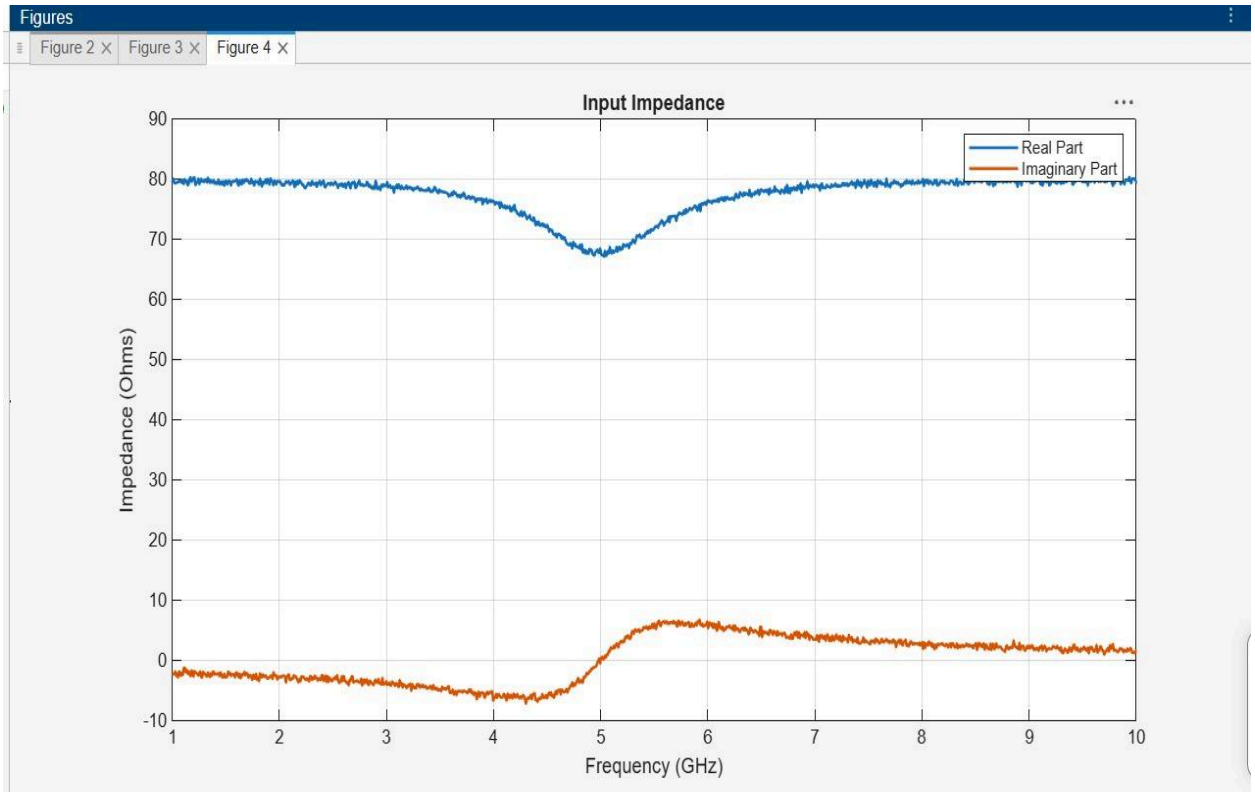

Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Approximate Antenna Gain
Estimated Gain (dBi)
6.5
6
5.5
5
4.5
4
1.8
2
2.2
2.4
2.6
2.8
3
Frequency (GHz)

Exp 13:Simulation-Based Frequency-Domain Characterization of a Microwave Two-Port Network Using S-Parameters and Network-Analyzer Emulation

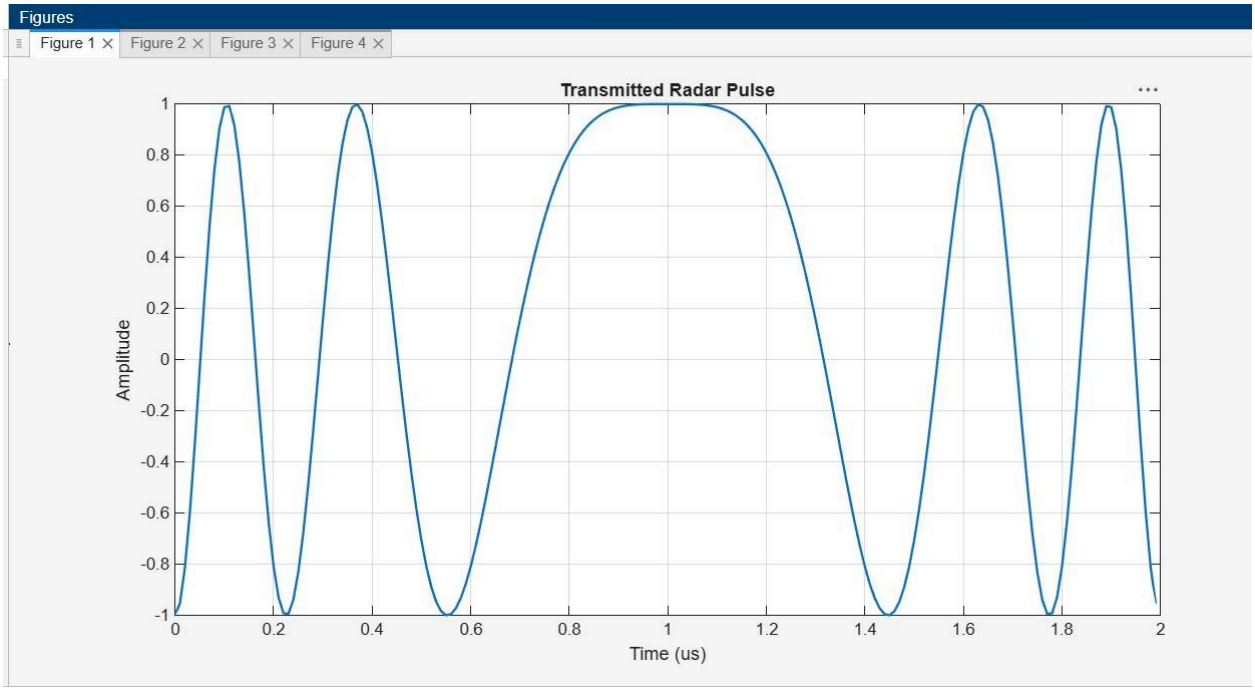


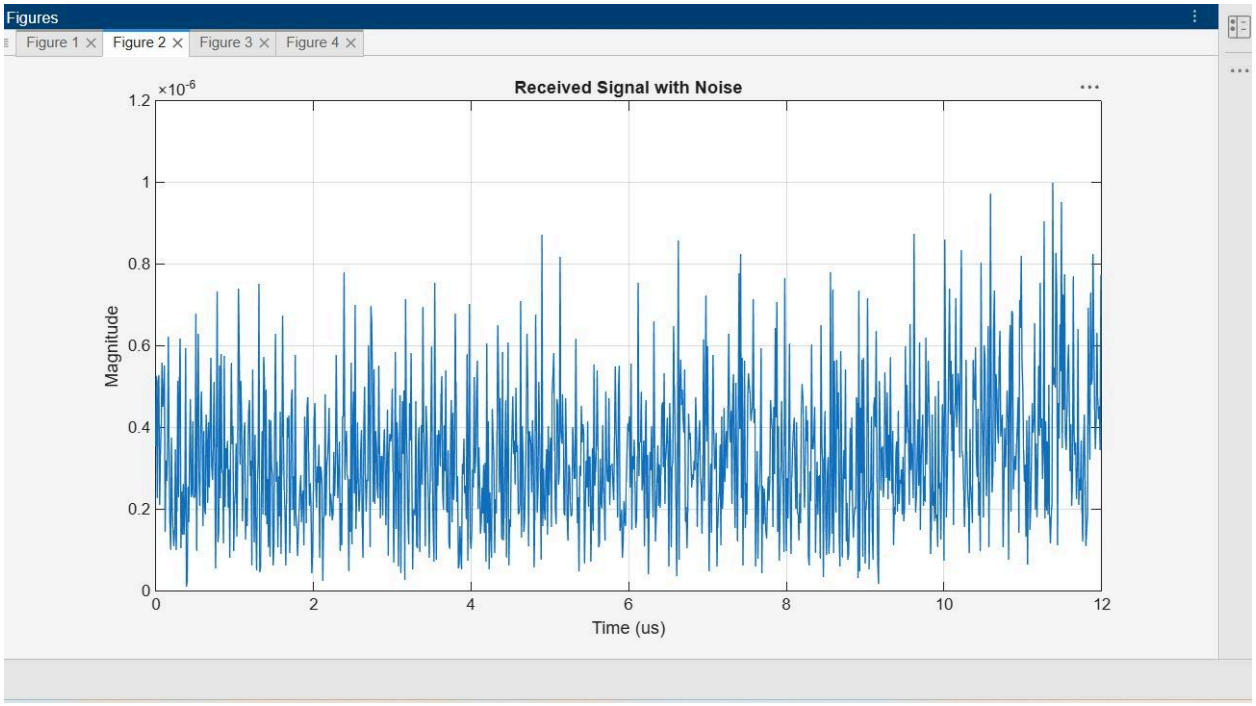


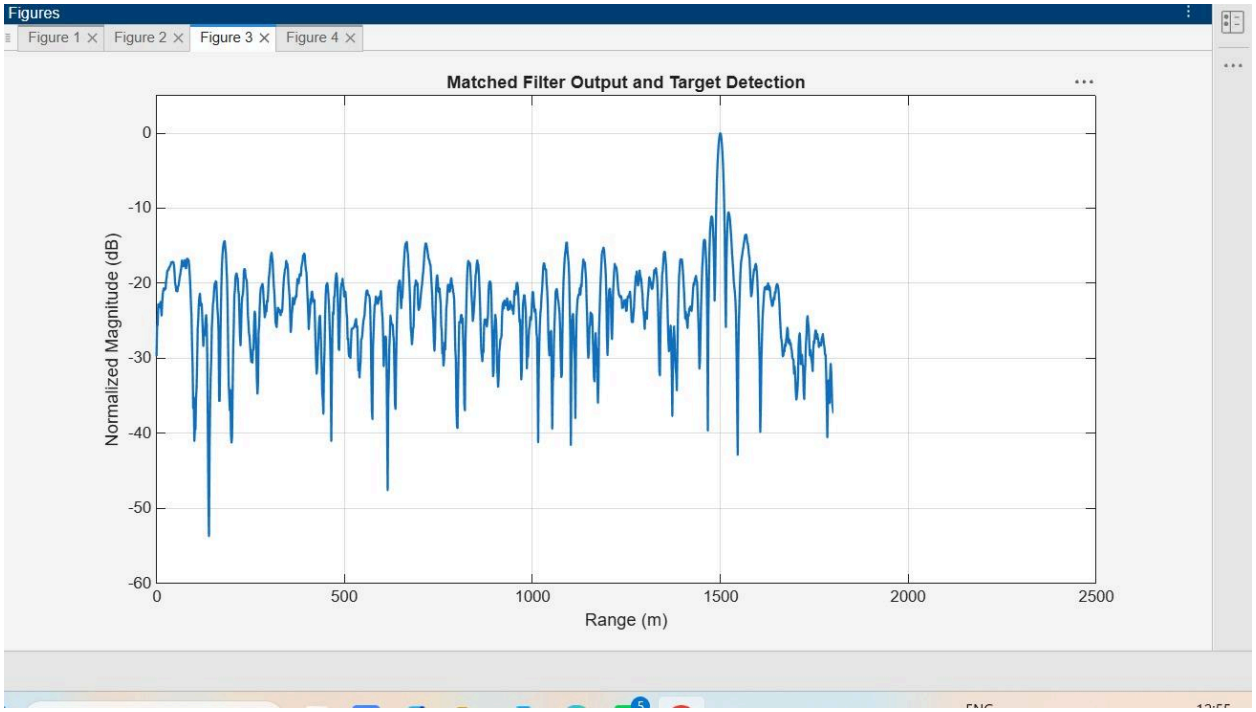

Figures
Figure 2
Figure 3
Figure 4
Voltage Standing Wave Ratio
VSWR
Frequency (GHz)
1.65
1.6
1.55
1.5
1.45
1.4
1.35
1.3
1
2
3
4
5
6
7
8
9
10

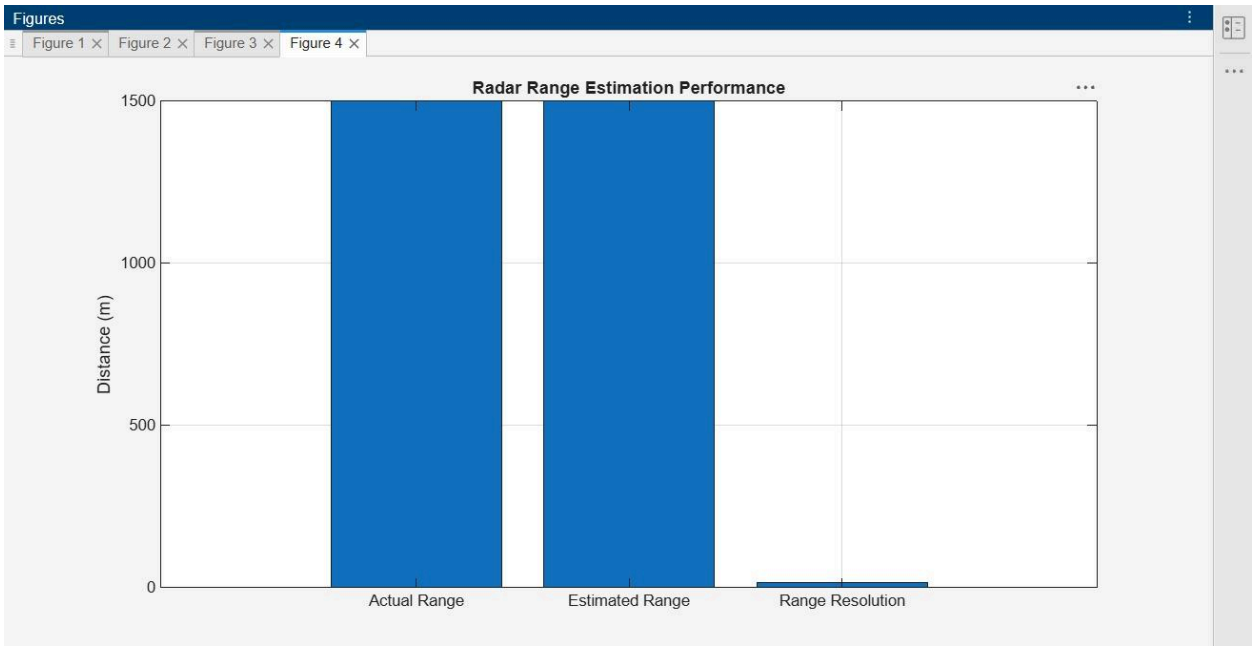

Figures
Figure 2
Figure 3
Figure 4
Input Impedance
Real Part
Imaginary Part
Impedance (Ohms)
Frequency (GHz)
90
80
70
60
50
40
30
20
10
0
-10
1
2
3
4
5
6
7
8
9
10

Exp 17:System-Level Simulation and Performance Analysis of a Microwave Radar for Target Detection, Range Estimation and Received-Signal Characterization






Figures
Figure 1
Figure 2
Figure 3
Figure 4
Matched Filter Output and Target Detection
Normalized Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
0
500
1000
1500
2000
2500
Range (m)


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Radar Range Estimation Performance
Distance (m)
1500
1000
500
0
Actual Range
Estimated Range
Range Resolution